

Project 1 Data Card

Title

Authors	Last updated	Created
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Card ID

Primary Dataset	<i>Adult</i>
Description	contains 48842 examples representing adult income from census bureau data in 1994.
Supplemental Dataset 1	<i>Countries of the World</i>
Description	<i>This dataset contains 227 examples relaying information about countries from CIA factbook data from 1970 - 2017.</i>

Datasets at a glance

Primary dataset: Adult

The primary dataset contains 48842 examples representing adult census data from census data in 1994.

Fields

Field name	Data Type	Description
age	Integer	Age of individuals (>16)
workclass	Categorical	Private, Self-emp-not-inc, Self-emp-inc, Federal-gov, Local-gov, State-gov, Without-pay, Never-worked.
fnlwght	Integer	This is a proxy for the demographic background of the people: "People with similar demographic characteristics should have similar weights". This similarity-statement is not transferable across the 51 different states.
education	Categorical	Bachelors, Some-college, 11th, HS-grad, Prof-school, Assoc-acdm, Assoc-voc, 9th, 7th-8th, 12th, Masters, 1st-4th, 10th, Doctorate, 5th-6th, Preschool.
education-num	Integer	Years of education
marital-status	Categorical	Married-civ-spouse, Divorced, Never-married, Separated, Widowed, Married-spouse-absent, Married-AF-spouse.
occupation	Categorical	Tech-support, Craft-repair, Other-service, Sales, Exec-managerial, Prof-specialty, Handlers-cleaners, Machine-op-inspct, Adm-clerical, Farming-fishing, Transport-moving, Priv-house-serv, Protective-serv, Armed-Forces.
relationship	Categorical	Wife, Own-child, Husband, Not-in-family, Other-relative, Unmarried.
race	Categorical	White, Asian-Pac-Islander, Amer-Indian-Eskimo, Other, Black.
sex	Binary	Female, Male.
capital-gain	Integer	Gains from the sale of assets.
capital-loss	Integer	Loss from the sale of assets.
hours-per-week	Integer	Working hours in a week
native country	Categorical	United-States, Cambodia, England, Puerto-Rico, Canada, Germany, Outlying-US(Guam-USVI-etc), India, Japan, Greece, South, China, Cuba, Iran, Honduras, Philippines, Italy, Poland, Jamaica, Vietnam, Mexico, Portugal, Ireland, France, Dominican-Republic, Laos, Ecuador, Taiwan, Haiti, Columbia, Hungary, Guatemala, Nicaragua, Scotland,

		Thailand, Yugoslavia, El-Salvador, Trinidad & Tobago, Peru, Hong, Holand-Netherlands.
income	Binary	>50K, <=50K.

Secondary dataset: Countries

The supplemental dataset contains 227 examples representing countries in the period from 1970 - 2017.

Fields (if applicable)

Field name	Data Type	Description
Country	Categorical	Name of country
Region	Categorical	Name of the region of the country
Population	Integer	Population of the country
Area	Integer	Square footage of the country
Population Density	Integer	Population relative to the area of the country
Coastline	Integer	Length of the coastline
Net Migration	Integer	Number of people moving into and out of the country
Infant Mortality	Integer	Number of infant deaths in the country annually
GDP per Capita	Integer	Average earning income relative to the population
Literacy	Integer	Percentage of the country that can read and write.
Phones (per 1000)	Integer	Number of phones per 1000 people
Arable	Integer	Percentage of land capable of handling crops
Crops	Integer	Percentage of land with crops
Other	Integer	Percentage of land used for purposes other than agriculture
Climate	Categorical	Average climate on a scale of 1 -25
Birthrate	Integer	Birthrate in the country
Deathrate	Integer	Death rate in the country
Agriculture	Integer	GDP sector composition - amount related to agriculture

Industry	Integer	GDP sector composition - amount related to industry
Service	Integer	GDP sector composition - amount related to service

Usage information

Use(s)

The primary data set allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

The secondary data set is only allowed to be used in the USA. Other than that, neither of the datasets have privacy related considerations or issues.

Licensing

License type for each data source and the license's corresponding requirements for sharing, transforming, logging changes, citing, and restricting usage of the data.

Primary: This dataset is licensed under a [Creative Commons Attribution 4.0 International](https://creativecommons.org/licenses/by/4.0/) (CC BY 4.0) license.

Secondary: CC0: Public Domain

Privacy considerations

The primary data set allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

The secondary data set is only allowed to be used in the USA. Other than that, neither of the datasets have privacy related considerations or issues.

Format

The primary dataset is in a CSV file.

The first supplemental dataset is in a CSV file

Source and collection process

Origin

Primary: Extraction was done by Barry Becker from the 1994 Census database (<https://www.census.gov/programs-surveys/ahs/data/1994.html>). A set of reasonably clean records was extracted using the following conditions: ((AGE>16) && (AGI>100) && (AFNLWGT>1)&& (HRSWK>0)).

Secondary: All these data sets are made up of data from the US government.
<https://www.cia.gov/the-world-factbook/countries/>

Original datasets author/publisher

Primary: Ronny Kohavi and Barry Becker
Secondary: Fernando Lasso

Original datasets point of contact

Name and contact information of source datasets point of contact.
Primary: ronnyk@live.com, ml-repository@ics.uci.edu
Secondary: N/A

Original metadata

Online link to original metadata file for your source datasets.
Primary: <https://archive.ics.uci.edu/dataset/2/adult>
Secondary: <https://www.cia.gov/the-world-factbook/countries/>

Collection process

Online link(s) to information on data source's collection and labeling processes.
Primary: <https://www.kaggle.com/datasets/uciml/adult-census-income>
Secondary: <https://www.kaggle.com/datasets/fernandol/countries-of-the-world/data>

Collection or access date

Primary:
Secondary: 01/30/2026

Refresh date/schedule

Primary: N/A
Secondary: N/A

Exploration

Questions driving the exploration study

1. How does the probability of earning more than \$50K differ when considering age groups, gender, and racial categories?
2. What is the relationship between education level and the likelihood of earning more than \$50K?
3. How do hours worked per week differ between income groups ($\leq$ \$50K vs $>$ \$50K), and does this relationship change based on the occupation of the individual?
4. Which occupations have the highest proportion of high-income earners, and how do these patterns differ by education level?
5. How does an individual's native country of origin relate to income, and how do these patterns change when comparing U.S.-born and non-U.S.-born individuals?
6. When grouped by region, do immigrants from different regions show different income patterns in?
7. How does changing the income threshold (e.g., \$40K, \$50K, \$60K) affect income disparity when comparing gender and race?

Descriptive statistics

Add basic statistics for each field here, as relevant, e.g. some statistics will be relevant for numeric data, but not for strings.

Primary dataset: Adult

	workclass	race	sex	native_country	income	education_num	hours_per_week
count	32561	32561	32561	32561	32561	32561.000000	32561.000000
unique	9	5	2	42	2	NaN	NaN
top	Private	White	Male	United-States	$\leq$ 50K	NaN	NaN
freq	22696	27816	21790	29170	24720	NaN	NaN
mean	NaN	NaN	NaN	NaN	NaN	10.080679	40.437456
std	NaN	NaN	NaN	NaN	NaN	2.572720	12.347429
min	NaN	NaN	NaN	NaN	NaN	1.000000	1.000000

25%	NaN	NaN	NaN	NaN	NaN	9.000000	40.000000
50%	NaN	NaN	NaN	NaN	NaN	10.000000	40.000000
75%	NaN	NaN	NaN	NaN	NaN	12.000000	45.000000
max	NaN	NaN	NaN	NaN	NaN	16.000000	99.000000

Secondary dataset: Countries

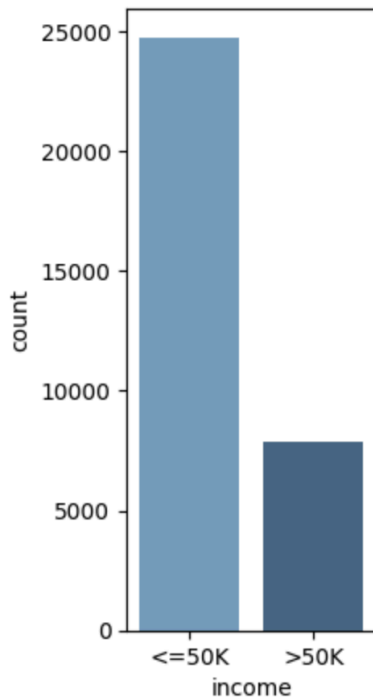
	population_d ensity	literacy	gdp_per_capi ta	agriculture	industry	service
count	227	209	226.000000	212	211	212
unique	219	140	NaN	150	155	167
top	13,8	99,0	NaN	0,01	0,17	0,684
freq	2	13	NaN	8	4	5
mean	NaN	NaN	9689.823009	NaN	NaN	NaN
std	NaN	NaN	10049.138513	NaN	NaN	NaN
min	NaN	NaN	500.000000	NaN	NaN	NaN
25%	NaN	NaN	1900.000000	NaN	NaN	NaN
50%	NaN	NaN	5550.000000	NaN	NaN	NaN
75%	NaN	NaN	15700.000000	NaN	NaN	NaN
max	NaN	NaN	55100.000000	NaN	NaN	NaN

Distribution plots

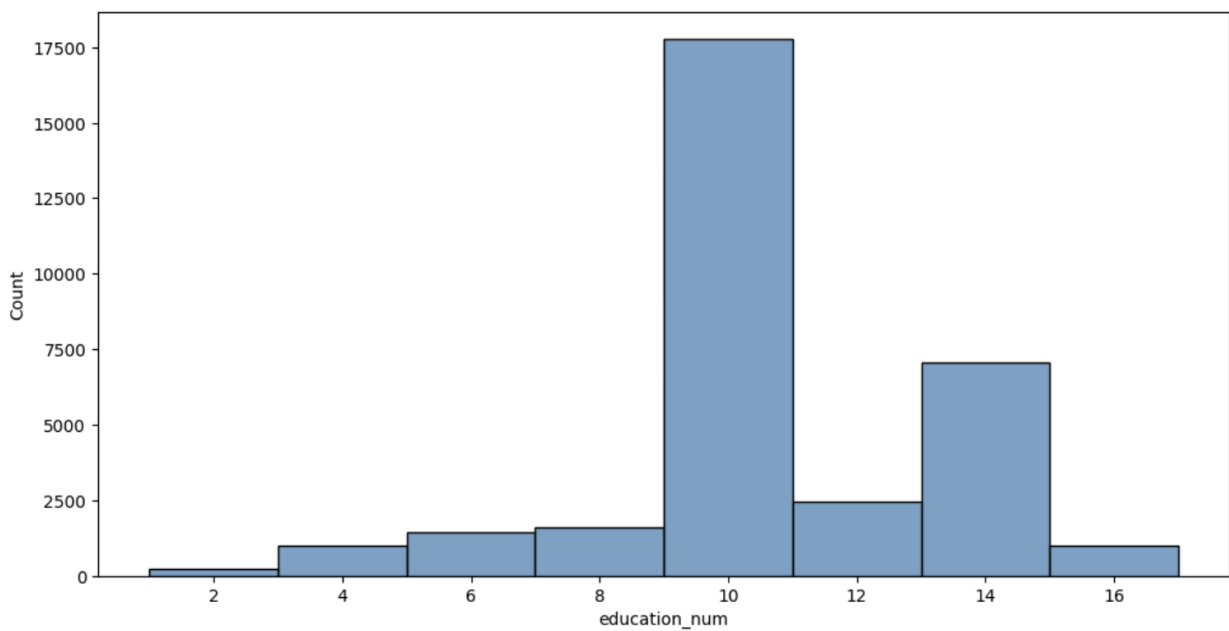
Include charts showing each key field's distribution.

Primary:

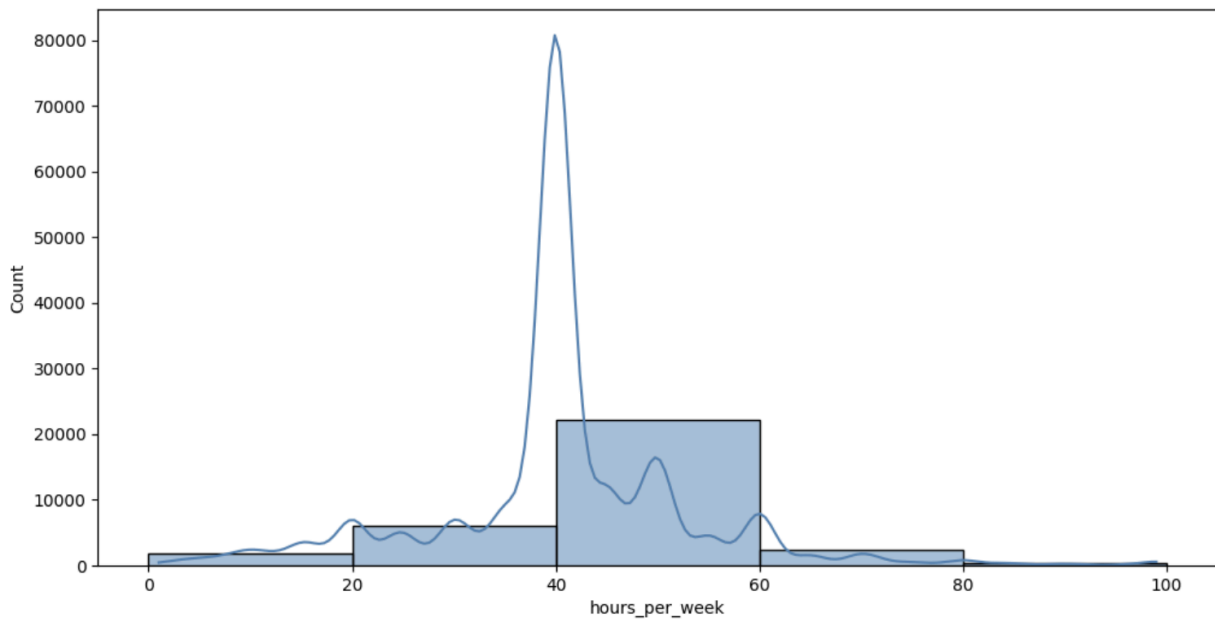
Income: This data is already categorized into 2 income ranges (equal to or below 50K and above 50k). People in the $\leq 50K$ category are 3 times more than those in the $>50K$ category, which shows that high earners represent only a minority of the population in the adult data set.



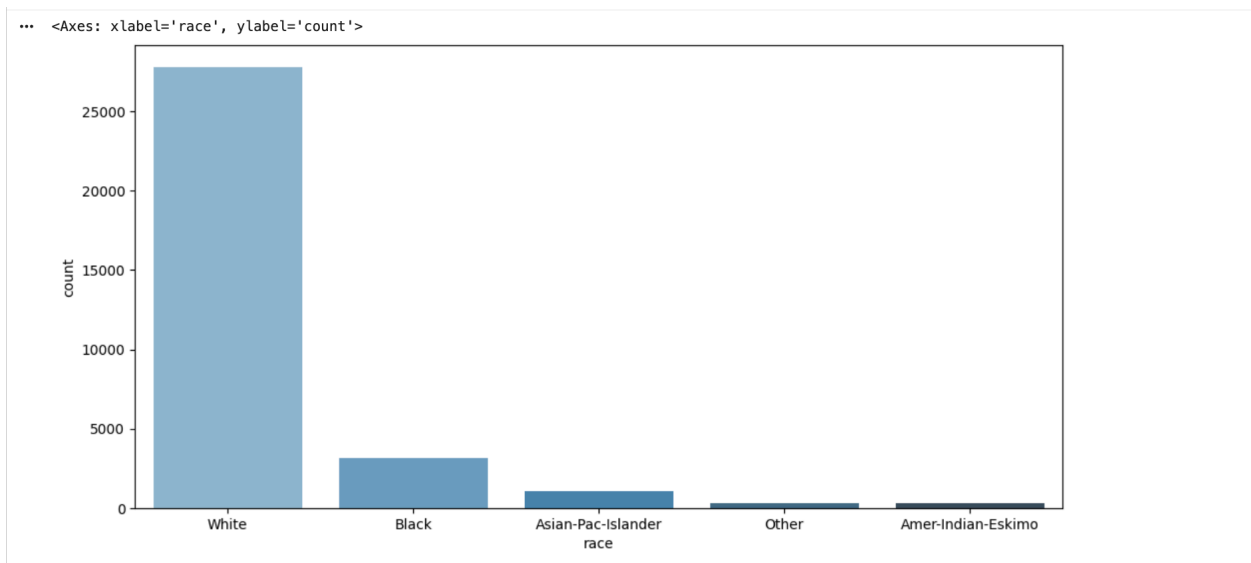
Years of Education: The years of education are grouped by range. This distribution is right-skewed with the majority of people's education ranging between 7-13 years. Moving past 11 years, however, the number of people declines significantly indicating that fewer people get higher levels of education.



Hours per Week: The hours worked by adults in the US per week are grouped by range. The distribution is right-skewed with a majority of people working between 20-40 and 40-60. Very few people work in the extreme ranges of below 20 and above 60 hours meaning most people work part-time to full-time.

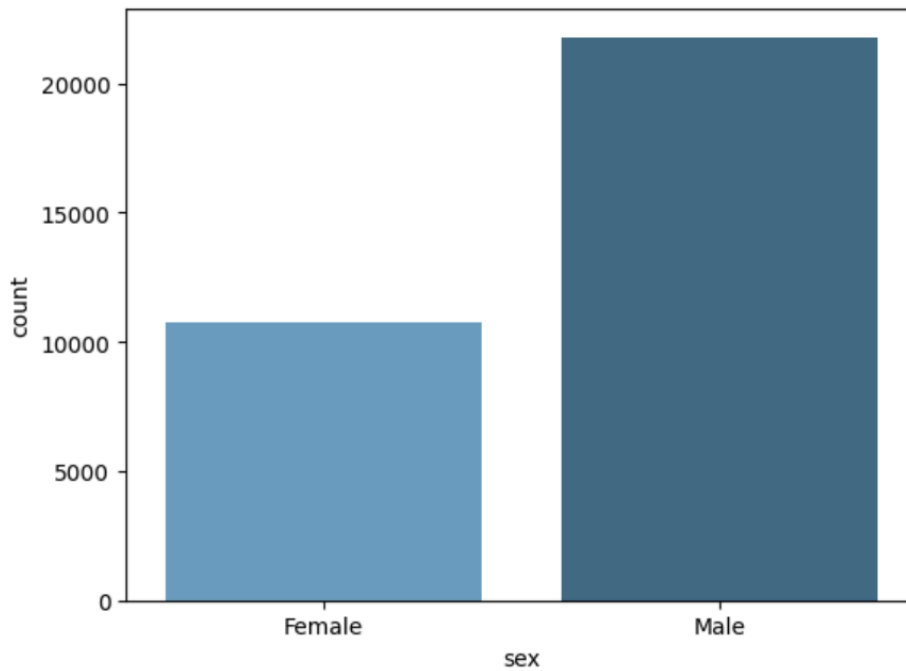


Race: Race categories are displayed by count. The distribution is really unbalanced and a majority of the entries in the data set identify as White. The other racial groups make up a very small part of the data set with most of the minority group identifying as black. This imbalance is important to keep in mind for the future because it could be an important consideration when drawing future conclusions.



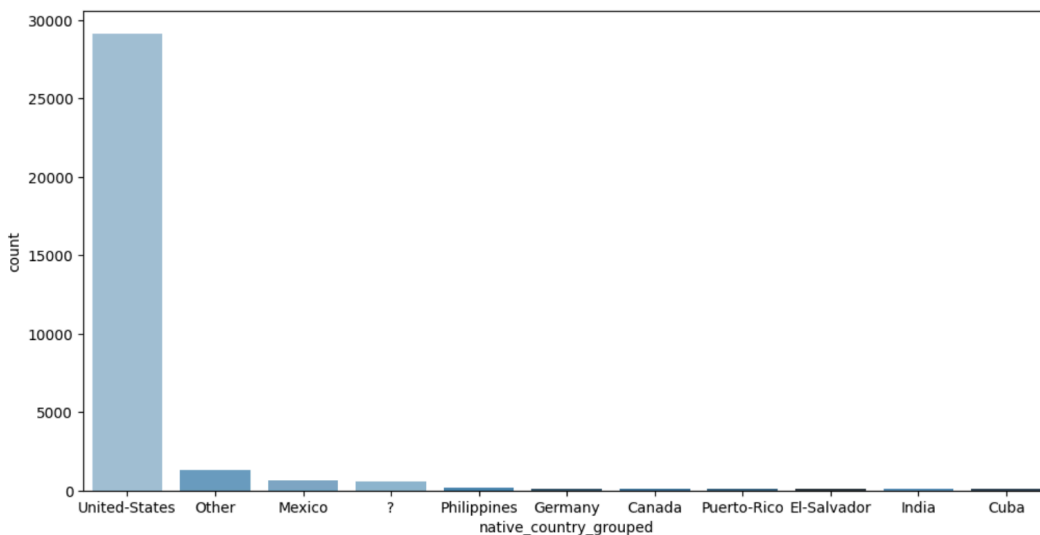
Gender: The two genders are displayed by count. The distribution heavily favors males with over half of the data set being male. Again, this is a notable observation to consider when we draw future conclusions.

```
... <Axes: xlabel='sex', ylabel='count'>
```



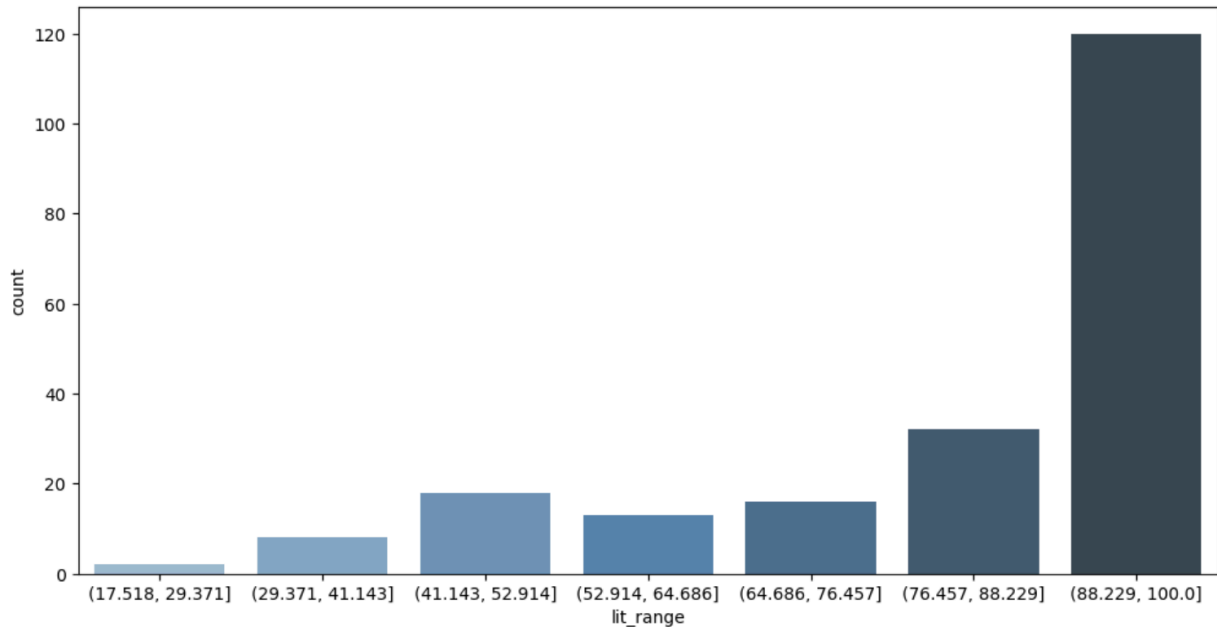
Native Country: This count plot highlights the top ten countries when it comes to adults from those countries with the rest of the countries being labeled as Other. This graph also shows an imbalance in the data when it comes to international representation in the data set. The majority of people in this data set are in the United States.

```
<Axes: xlabel='native_country_grouped', ylabel='count'>
```

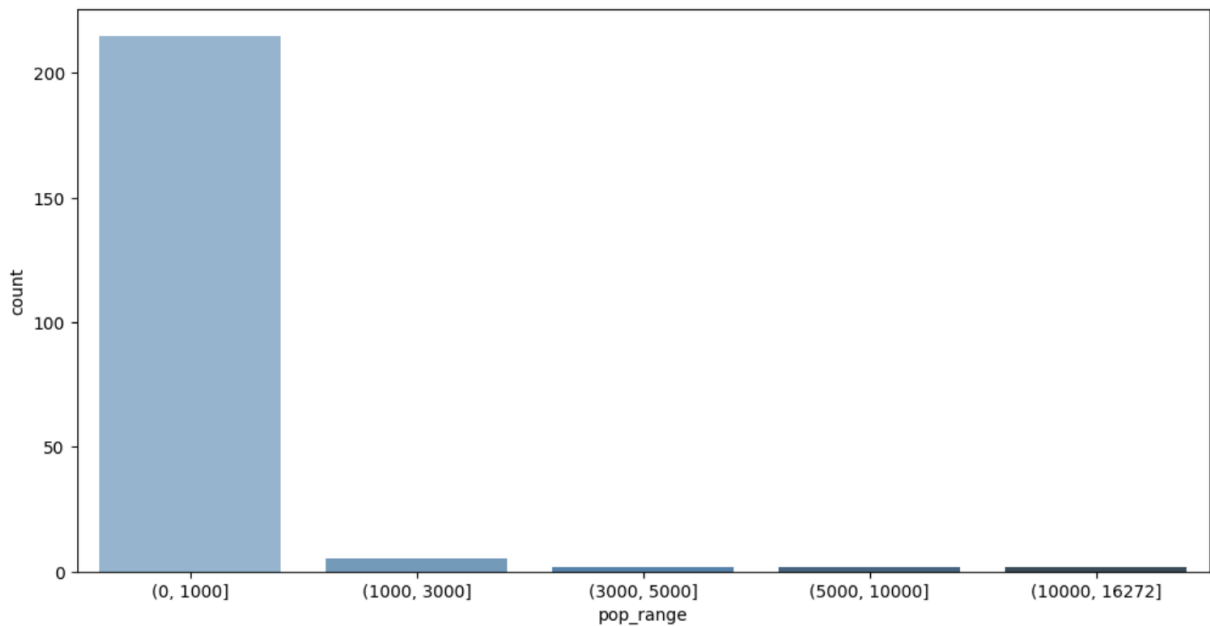


Secondary:

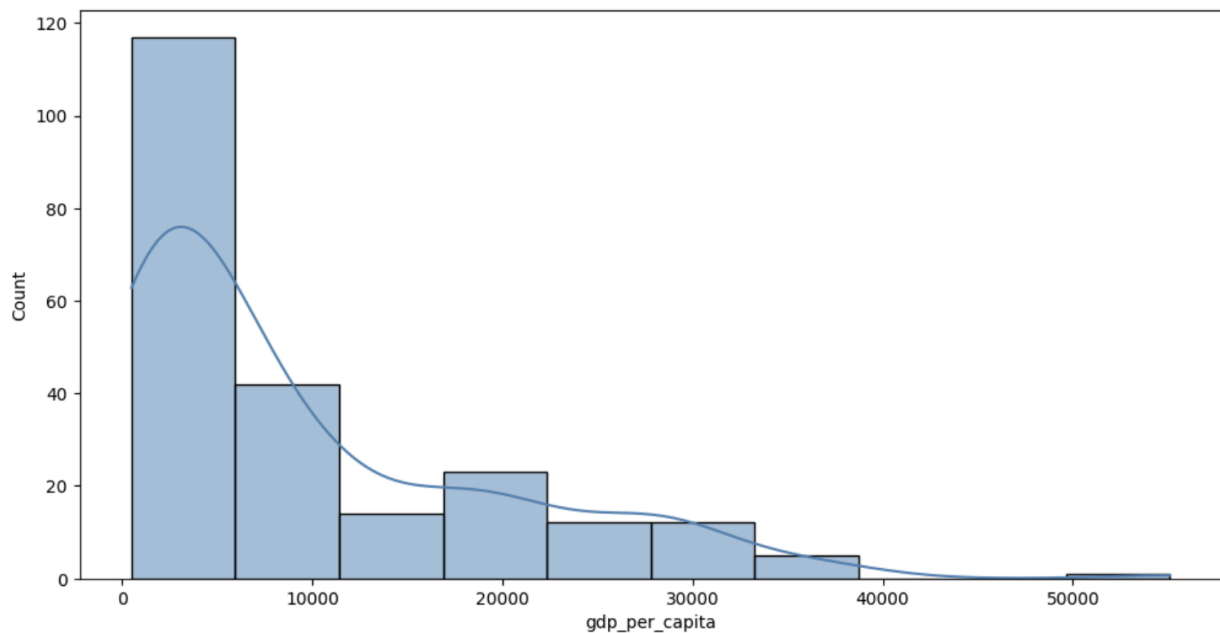
Literacy: Literacy percentage is grouped by range. This plot is left-skewed, where the highest concentration is in the 88%-100% range. Most countries have a high literacy rate, a very few are at the extremely low end.



Population density: population density grouped by range. The most popular population density lies between 0 and 1000. With outliers lying around 3000-16272, this can be caused by countries with a small land area that have a moderately large population.



GDP per capita: GDP per capita grouped by range. This plot is right-skewed, and the wealth gap between countries is evident, with most countries concentrated in the lowest income range (0-10000). The decrease in the number of countries with higher GDPs suggests a global wealth disparity.



Visual analysis

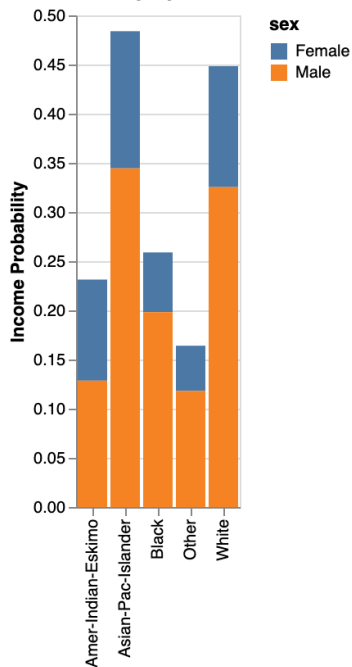
Question 1: How does the probability of earning more than \$50K differ when considering age groups, gender, and racial categories?

Data Nature:

- gender: categorical (binary)
- race: categorical
- income: binary (converted to probability)

Grouped Bar Chart (income probability v. gender + race) - Both gender and race are categorical variables and the grouped bar chart will allow for a direct comparison of having an income higher than \$50K across the categories. Through the graph we hope to see if gender gaps when it comes to pay exist between racial groups.

Income Probability by Race and Gender ...



Insights

- Males have a significantly higher income probability across all of the racial groups, indicated by the higher bars in every age group. This is a clear representation of the gender income disparity that is prevalent regardless of a person's race.
- Asian-Americans and Pacific Islanders have the highest overall income probability. Males have the highest probability of around 0.34 of being above \$50K and females also have a relatively high probability when compared to the other racial groups. They have the highest representation in the higher-income bracket.
- White individuals also have a high income probability, with the males being pretty close behind Asian-Americans with a probability of around 0.32.
- The Black, Other, and American-Indian categories all show significantly lower income probability compared to the White and Asian_american groups. This indicates not only a gender disparity, but also a significant disparity between the various racial groups.
- When it comes to the actual gender gap, it is the largest among White and Asian-American groups, so even though they have higher income probabilities, the actual income gap between gender in these racial groups is higher.

Question 2: What is the relationship between education level and the likelihood of earning more than \$50K?

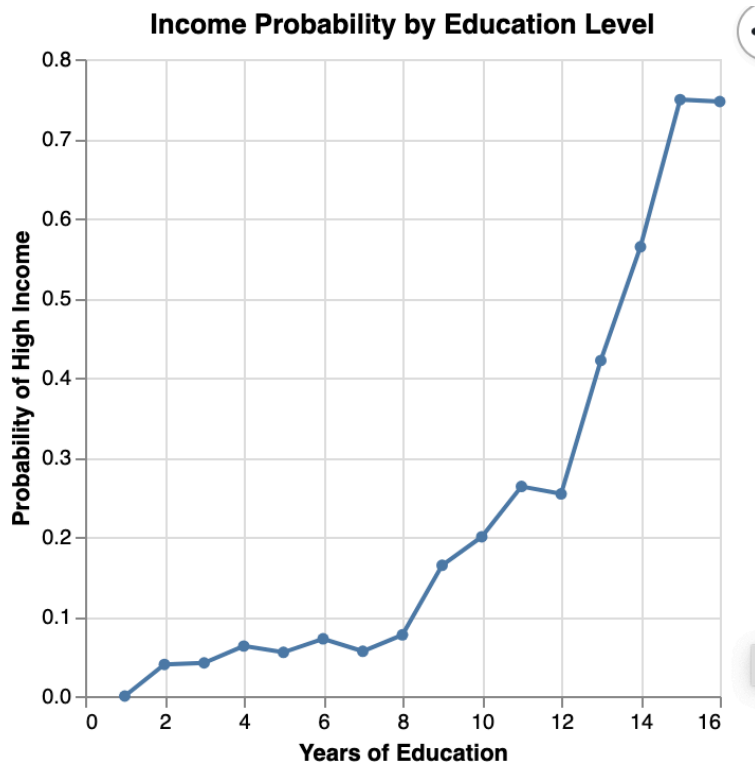
Data Nature:

- education_num: ordered numeric
- income: binary (converted to probability)

Chart Mapping

1. Line Chart - This chart is going to show income probability by years of education. Education is ordinal and represents an individual gaining skills over time. The line chart will visualize this gain

in skills over time and show the retention/benefit (measured through the probability of earning a higher income) per year of additional income. We are aiming to see whether the income probability increases steadily or if there are significant jumps at certain educational thresholds.



Insights

- Income probability increases very strongly with education. From 1-8 years the income probability stays pretty low. This makes sense because this is the group of people that have not completed high school. Starting around 9-10 years the probability increases a little more noticeably because they have some high school education. The jump in probability becomes very noticeable after 12 years of education because at this point people have a high school diploma. From there, the probability continues to grow significantly faster than from 1-12 years.
- There are specific thresholds when it comes to jump in probability that somewhat aligns with the education cycle. From 8-10 years the probability doubles. Then from 10-12 the probability grows by around 1.5 times. Finally from 12-15, the probability grows at a pretty steady rate at close to 2 or greater than 2 times per year.
- As expected, the lower education years have a probability very tightly clustered between 0 and 0.1 with very little upward movement. This means that there is very little income mobility at lower education levels. However, after 12 years, the slope becomes steeper meaning that education has disproportionately large effects on earning a high-income.
- Finally, there is a slight plateau between 15 and 16 years indicating that there might be diminishing marginal returns after around 15 years of education. This brings up the possibility that after 16 years adding more and more years might not dramatically increase the probability of earning a higher income.

Question 3: How do hours worked per week differ between income groups, and does relationship change based on the occupation of the individual?

Data Nature:

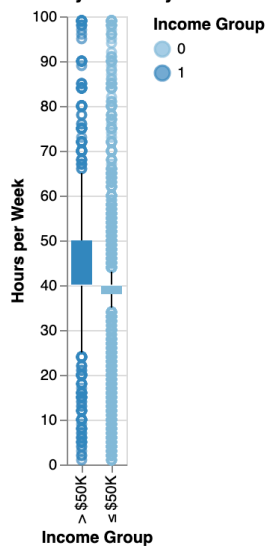
- hours_per_week: continuous numeric
- income: binary (converted to probability)
- occupation: categorical

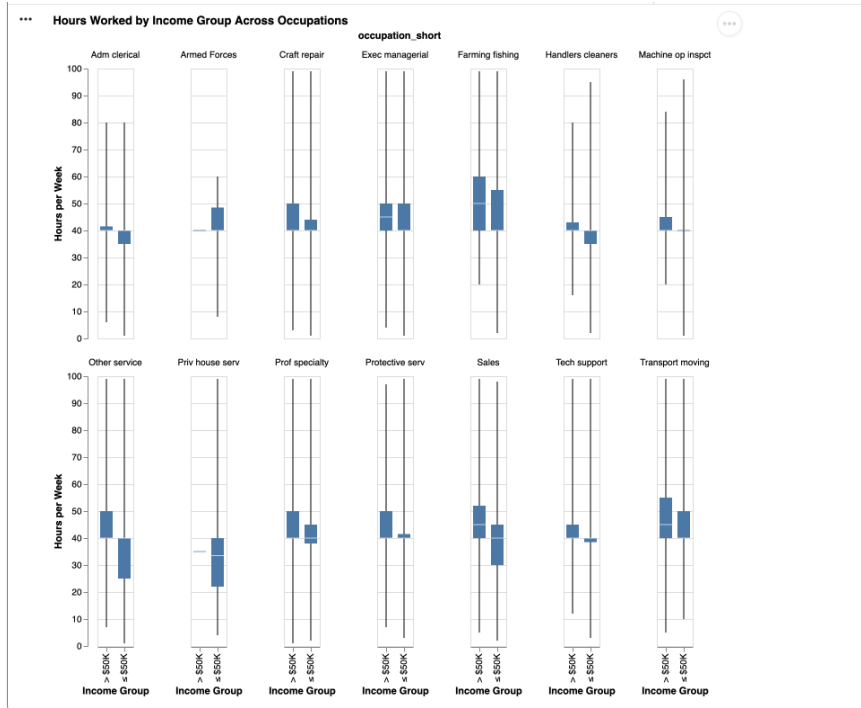
Chart Mapping

1. Boxplot (Hours by Income Group) - Boxplots summarize the median, quartiles, and spread of the data, so this will allow for comparison of the distribution overlap between the two income groups. The main goal behind this graph is to see whether high earners consistently work more hours or if the distributions of the hours worked overlap significantly.

2. Faceted Boxplots (Hours by Income within Occupation) - Faceting the boxplots helps to visualize whether occupation moderates the relationship that we established between the hours worked and the income group. The main goal behind this graph is to add on to the observations made in the previous plot to see if the occupation structure helps to better explain the relationship between hours worked and income.

Distribution of Weekly Hours by Income Group



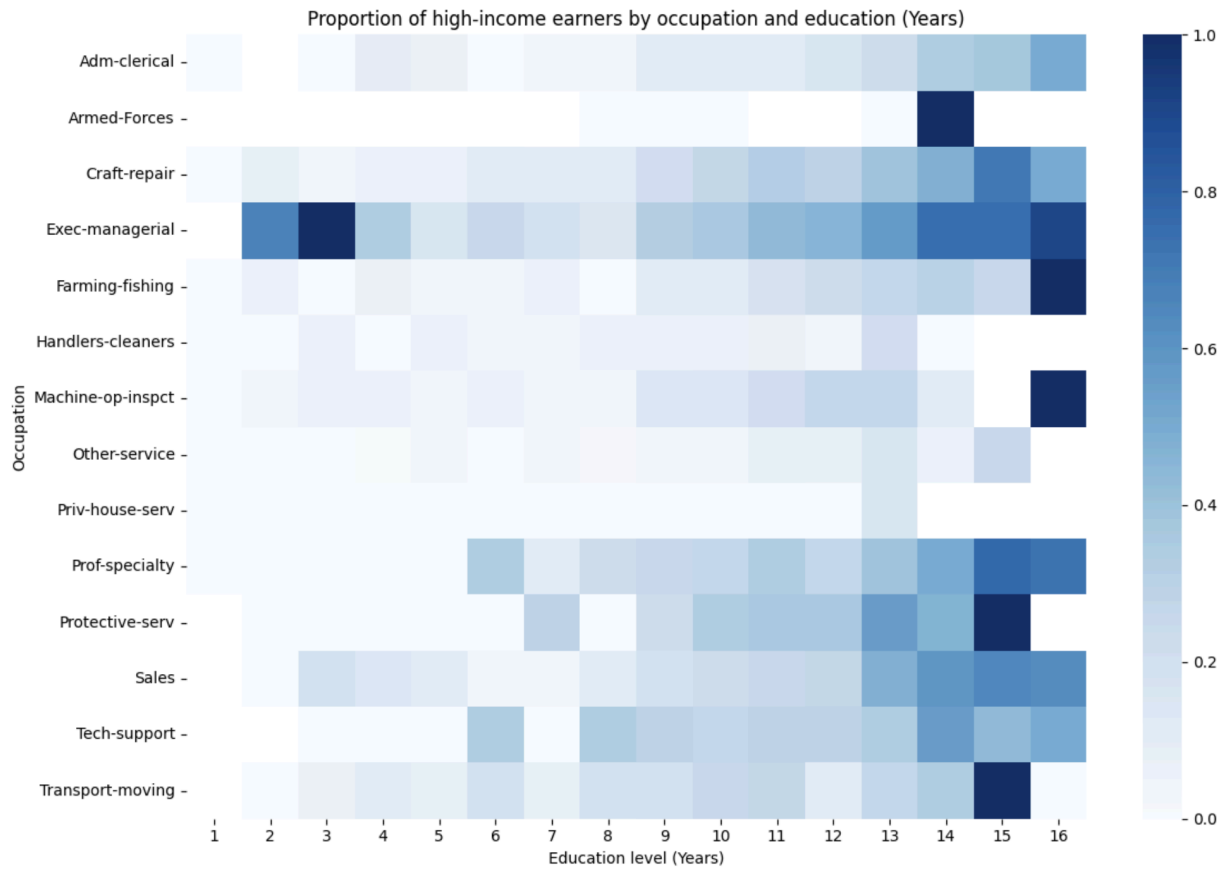


Insights

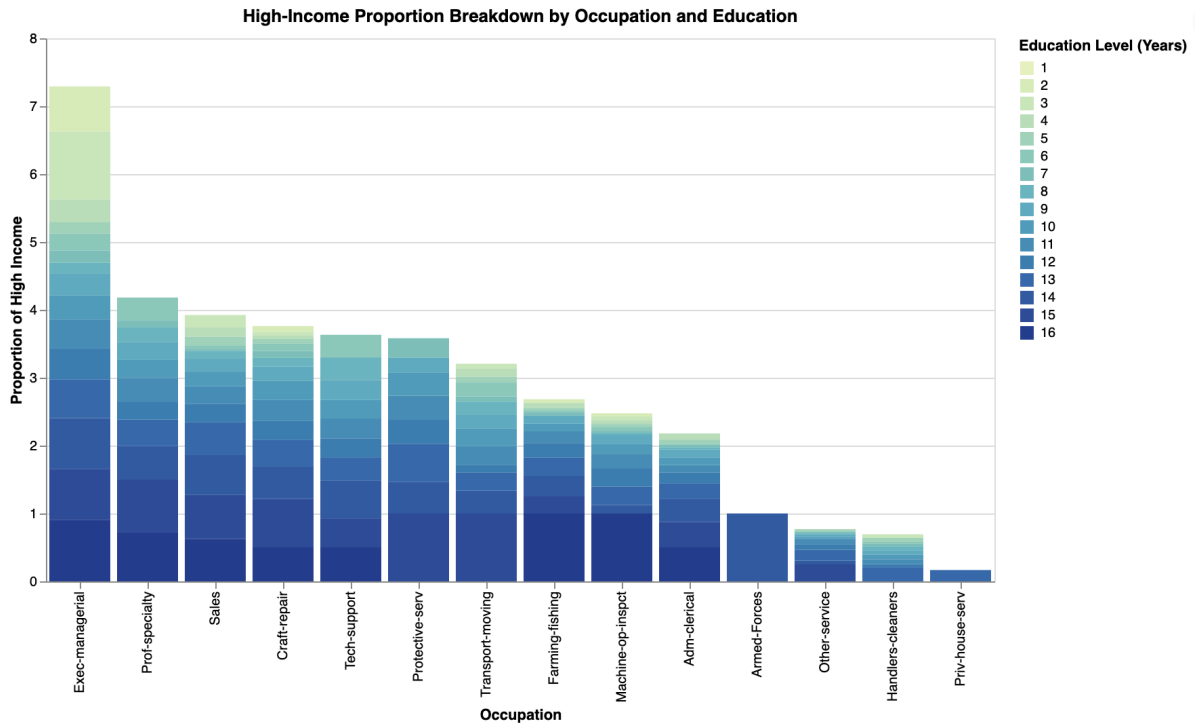
- Higher income is directly related with longer working hours at the aggregate level, indicated by the >\$50K group showing a higher median and upward-shifted distribution. However, the overlap between income groups indicates that hours worked alone is not the only determining factor of income.
- The relationship between hours and income is clearly dependent on occupation at very significantly. In fields such as executive-managerial, sales, and transport-moving, higher-income individuals lean towards working noticeably longer hours, indicating labor intensity contributes more strongly to income differences in these roles.
- In several occupations, increased hours do not really translate to proportionally higher income. Roles such as other-service and handlers-cleaners show limited upward income mobility despite long working hours, indicating structural wage ceilings or occupational constraints.
- Occupational structure seems to be a stronger determinant of income than just the hours worked. Other factors such as fields and skill level play a more significant role in shaping income outcomes.

Question 4: Which occupations have the highest proportion of high-income earners, and how do these patterns differ by education level?

Heatmap: it handles two categorical variables (Education and Occupation) simultaneously. It allows for a quick "pattern match" where the intensity of the color immediately reveals which combinations produce high earners



Stacked bar charts: By stacking the education levels, we can see not just which job pays more, but also how much of that total is contributed by different education tiers.



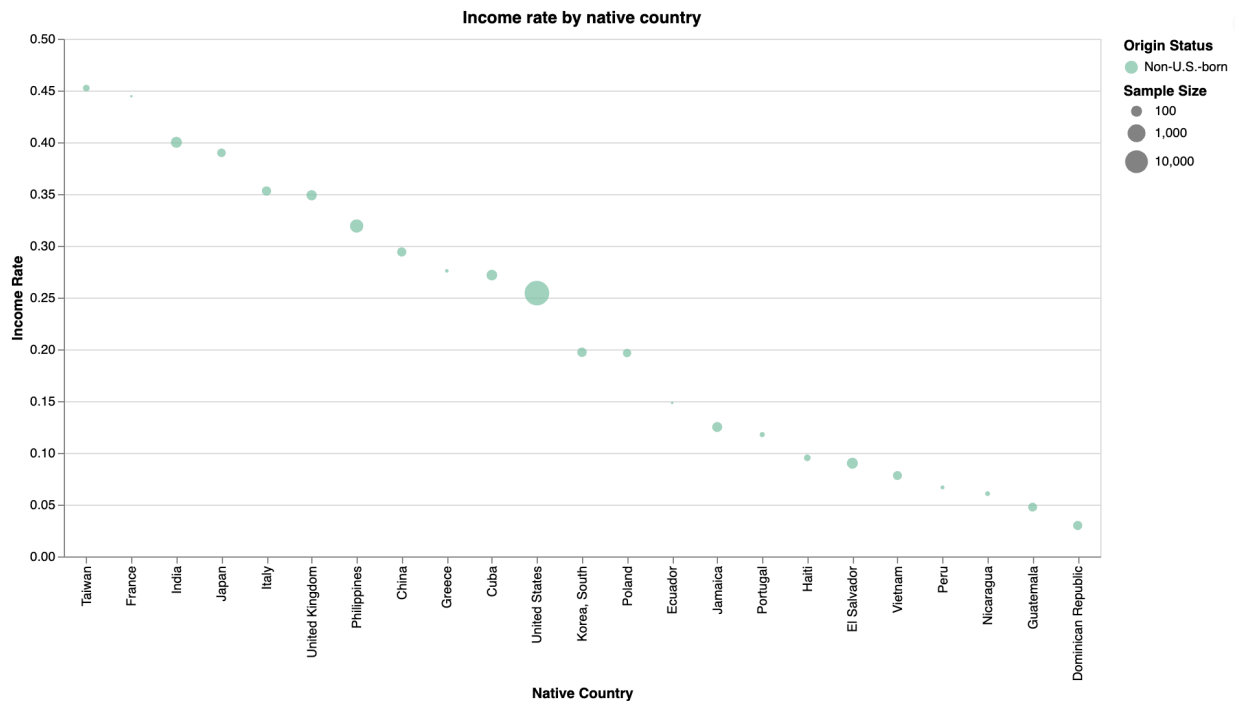
Based on the 2 figures, we can see that occupations with highest high-income proportions are exec-manage, prof-specialty, and sales.

The patterns also differ by education level:

- The general trend is that income would typically increase with years of education.
- Most occupations (especially protective-serv, craft-repair, and sales) see their most dramatic increase in high-income representation at exactly 14 and 15 years of education.
- Exec-manage is unique because it shows a moderate proportion of high earners even at mid-range education levels (3-5 years), though it still peaks significantly at 16 years of education.
- For Prof-specialty and Tech-support, the proportion of high-income earners is negligible at low education levels (1-8 years) but spikes once reaching higher education (13-16 years).
- However, roles like Handlers-cleaners, Other-service, and Priv-house-serv show very low proportions of high-income earners regardless of the years of education completed.

Question 5: How does an individual's native country of origin relate to income, and how do these patterns change when comparing U.S.-born and non-U.S.-born individuals?

Bubble plot: ideal for comparing individual countries because it displays 3 features: the country name, the income rate, and the sample size. This prevents misleading conclusions by showing that while 1 country might have a high income rate, its sample size is tiny compared to the other countries.



When comparing origin status, the data reveal significant differences in income distribution:

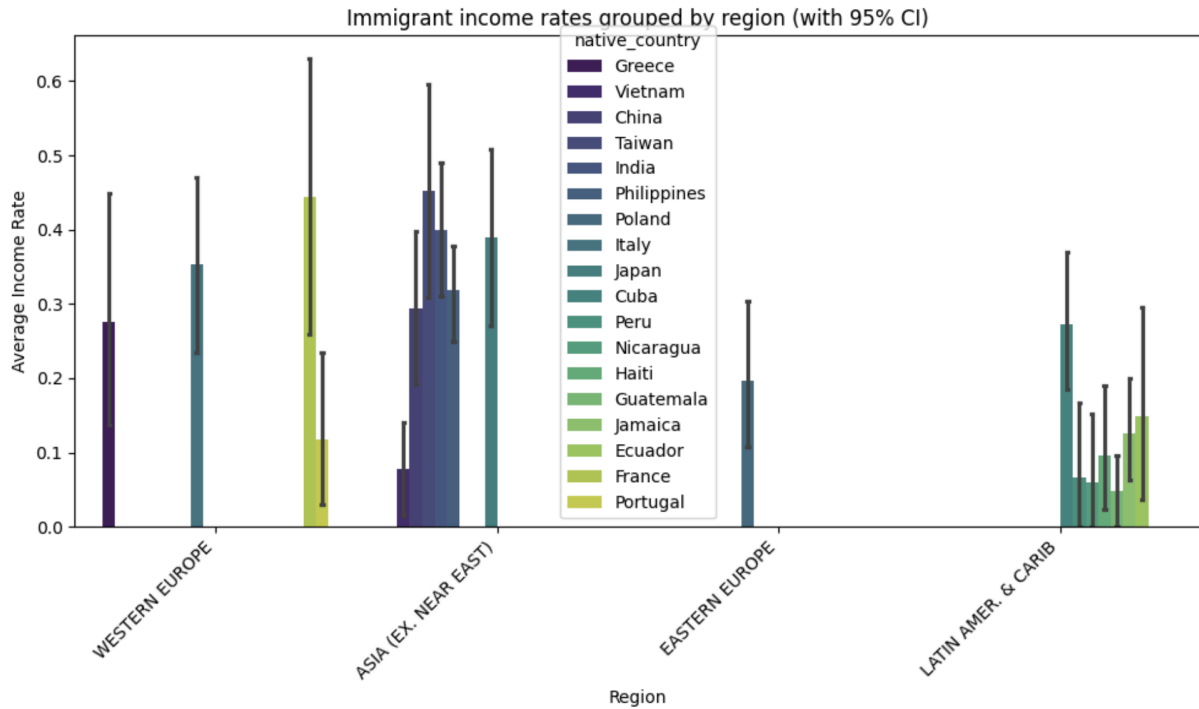
- Non-U.S.-born individuals represent both the highest and lowest ends of the income spectrum. For example, individuals from Taiwan and India (non-U.S.-born) significantly outperform the average U.S.-born income rate.
- The United States (U.S.-born) has a much larger sample size (largest bubble) compared to all other nations, yet its income rate (~0.25) is surpassed by several non-U.S.-born groups.

An individual's native country also shows correlations with their income rate:

- Highest: Taiwan (0.45), France, and India (0.40) represent the top-earning countries of origin.
- Mid-Range: Countries like the United Kingdom, Japan, and the United States (~0.25) occupy the middle of the spectrum.
- Lowest: individuals from Guatemala and the Dominican Republic has the lowest income rate, falling below 0.05.

Question 6: When grouped by region, do immigrants from different regions show different income patterns?

Bar chart with error bars: we use the bar chart to compare categorical means. We also include the confidence interval (CI) bars to show the reliability of the data.

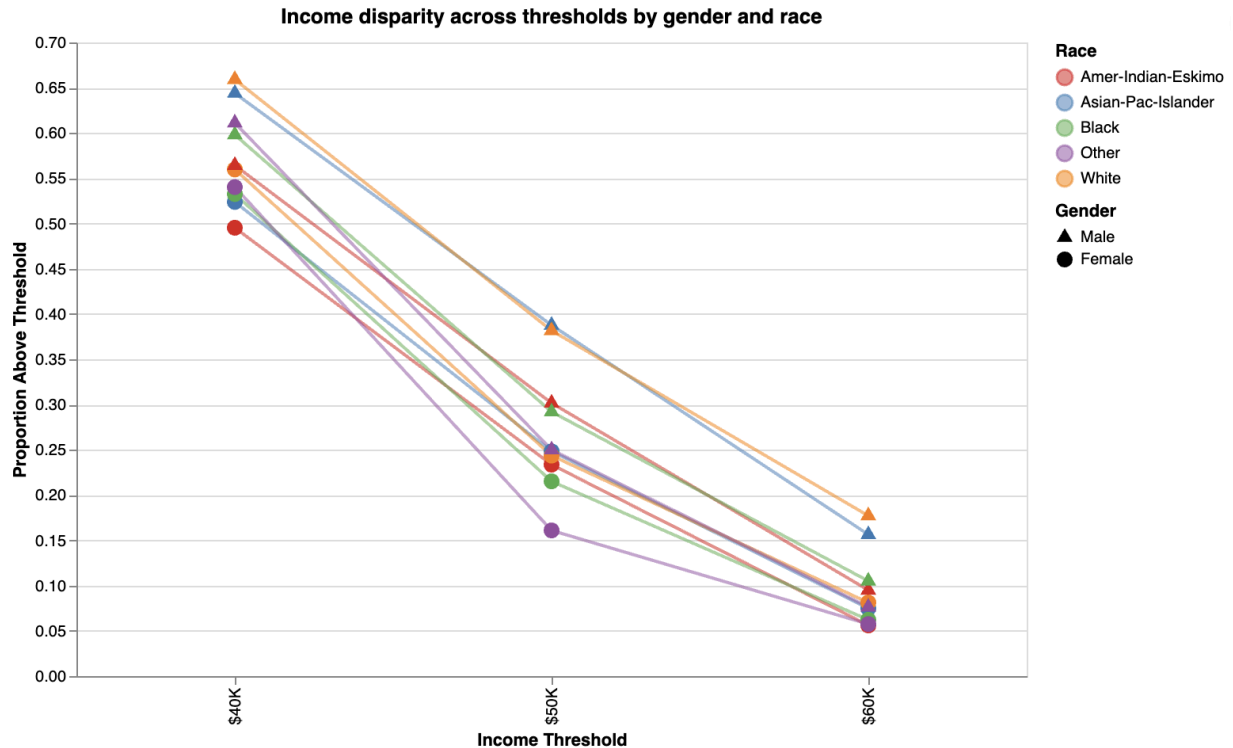


The data reveals a clear hierarchy in average income rates across the four analyzed regions:

- Western Europe & Asia (Ex. Near East): These regions generally display the highest average income rates but also show a wide range of income rates among countries in the same region. France (within Western Europe) and Taiwan and India (within Asia) stand out as top performers with average rates approaching or exceeding 0.40. Shows the highest internal disparity, with average rates ranging from below 0.10 (Vietnam) to over 0.45 (Taiwan). Western Europe also displays a wide gap, notably between the high-performing France and the significantly lower Portugal.
- Eastern Europe: This region shows a mid-range income pattern, largely driven by Poland, which has an average income rate of approximately 0.20.
- Latin Amer. & Carib: This region has the lowest income patterns among the groups. Most countries in this category, such as Nicaragua and Guatemala, show average income rates below 0.10.

Question 7: How does changing the income threshold (e.g., \$40K, \$50K, \$60K) affect income disparity when comparing gender and race?

Multi-line chart: we want to show trends across a scale. By plotting the proportion of high earners across 40K, 50K, and \$60K, the lines would reveal how income for different races and genders diverges at different thresholds.



Based on the line chart, increasing the income threshold significantly widens the disparities between groups.

Gender: The chart reveals a persistent and widening gap between genders across every threshold.

- At the \$40K threshold, the highest-performing male groups (White and Asian-Pac-Islander) sit between 0.65 and 0.66, while their female counterparts are significantly lower, between 0.53 and 0.54.

- As the threshold increases to \$60K, the proportion of females above the line drops sharply to approximately 0.06 to 0.09, while the top male groups maintain a much higher proportion near 0.16 to 0.18.

Race: disparities in this category are more pronounced as the income requirements are more stringent.

- Asian-Pac-Islander and White individuals (represented by blue and orange lines) consistently maintain the highest proportions across all thresholds.

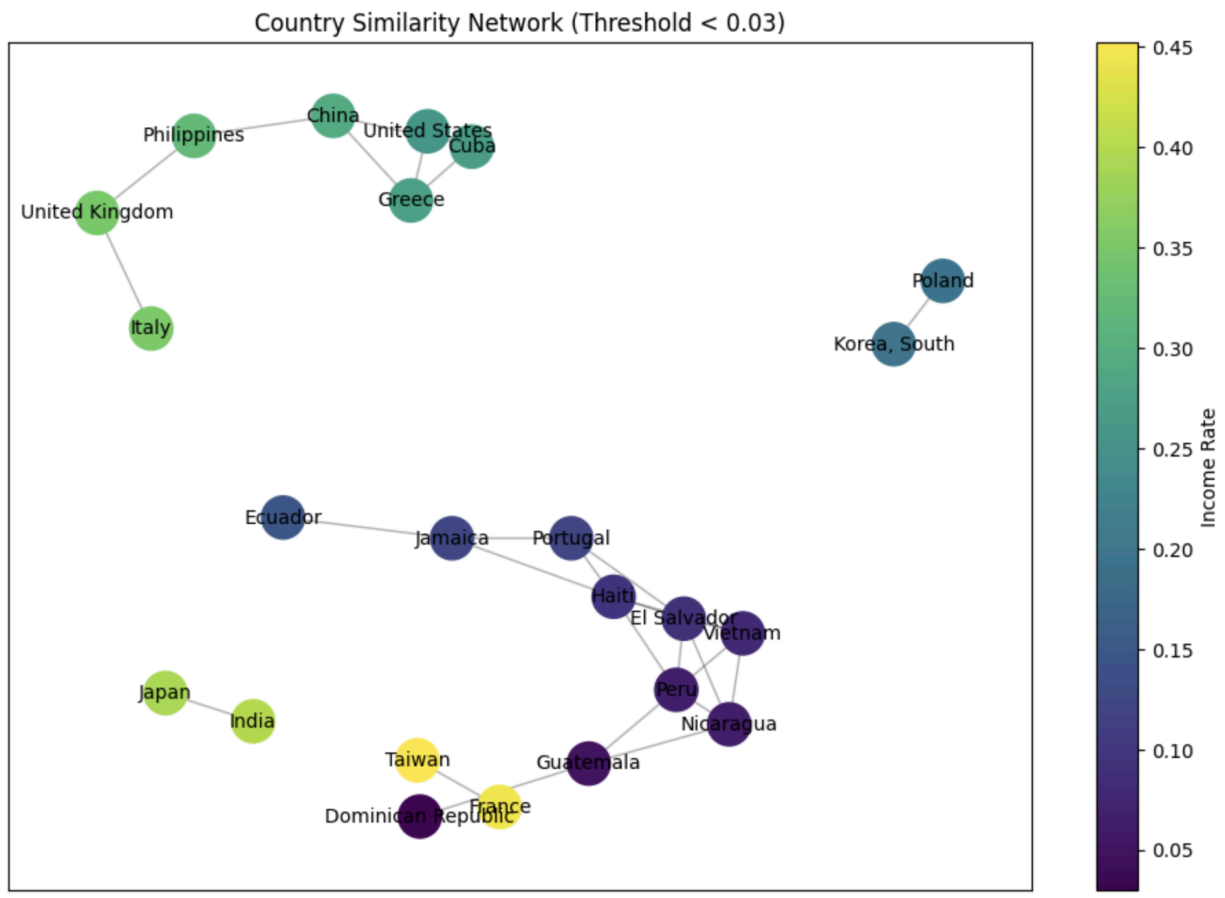
- Individuals identified as Other (purple) and Amer-Indian-Eskimo (red) experience the steepest declines. The "Other" female group drops from nearly 0.54 at \$40K to the lowest overall position of roughly 0.05 at \$60K.

- At the \$60K mark, most racial groups (excluding White and Asian males) converge at a very low proportion (below 0.11), suggesting that reaching the highest income tier is significantly difficult for these demographics, regardless of gender.

Analytical & statistical methods

Summarize your chosen quantitative methods and describe the specific insights or patterns they revealed about your data.

We use networkX with each node representing a country, it's colored based on a scale from low (purple) to high income rate (yellow), and the edges connect the countries whose income rates are within 3% of each other.



This network graph reveals that economic similarity creates a global web that goes beyond traditional geographic boundaries. By connecting countries within a 3% income rate threshold, several key properties are:

- The graph naturally shows wealth disparity among countries by showing distinct "tiers" where higher-income nations like France, Taiwan, and Japan are in the margin of this network, showing they have fewer economic neighbors within a 3% gap compared to the mid-income majority.
- A dense cluster exists among lower-to-mid income countries—such as Peru, Nicaragua, El Salvador, and Vietnam. This indicates they share a very narrow and consistent income range.

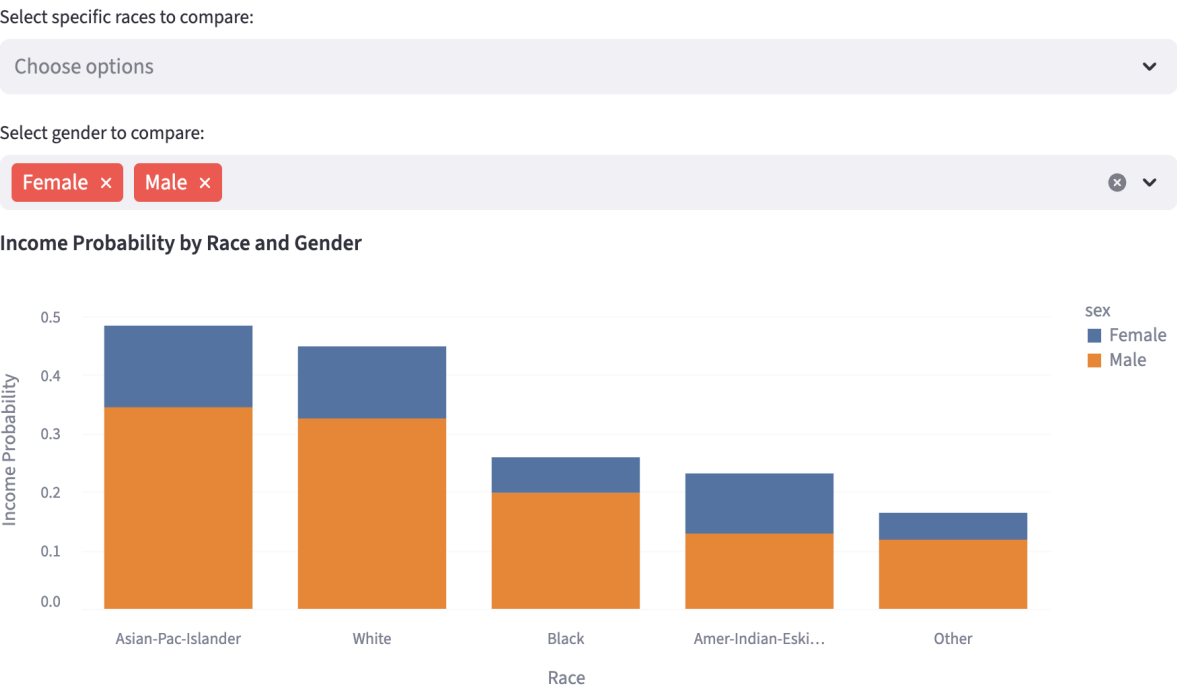
- The graph also reveals that nations with vastly different geographic and political contexts, such as the United States, China, Greece, and Cuba, actually share nearly identical income rates in this dataset.

Polished visualizations

Include the charts you refined and list the specific textbook design best practices applied.

1. Grouped bar chart

How does the probability of earning more than \$50K differ when considering gender and racial categories?



Visual Clarity: Removed outer chart borders and reduced gridline opacity to keep visual emphasis on the bars rather than background elements.

Improved Comparability: Sorted racial categories by overall income probability, allowing for identification of highest and lowest earning racial groups.

Strategic Highlighting: Applied the Tableau10 color palette for clearer gender differentiation and improved accessibility.

Accurate Scaling: Ensured the y-axis begins at zero and added slight padding above the maximum value to prevent compression maintain perception accuracy

4. Stacked bar chart

Demographics Exploration

Which occupations have the highest proportion of high-income earners, and how do these patterns differ by education level? 🔥

Select Education Year Range:

1

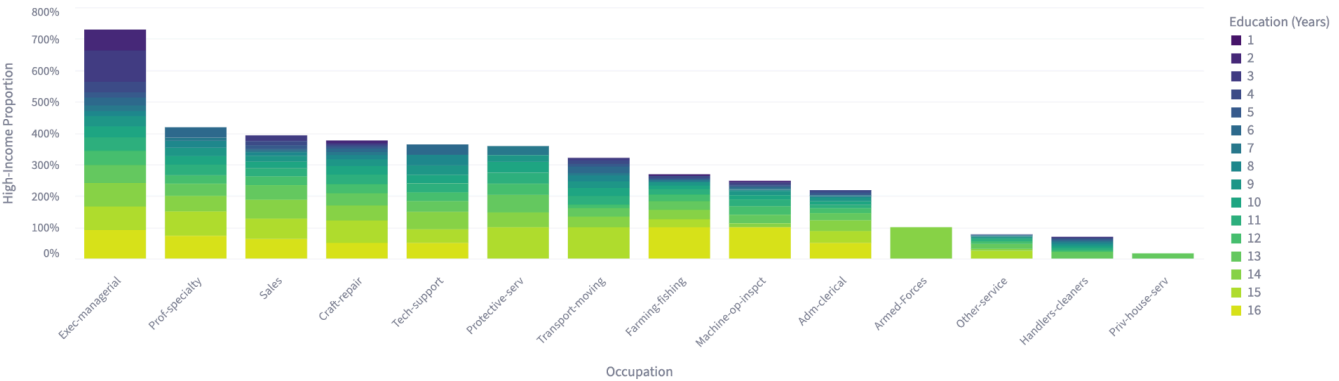
16

116

Filter by Occupation (Leave empty to show all):

Choose options

High-Income Proportion Breakdown by Occupation and Education
Visualizing the impact of educational attainment on earnings across different sectors

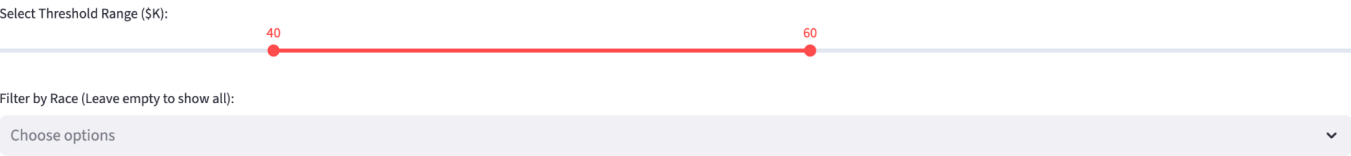


- Maximized data-ink ratio:** Removed the outer border and X-axis gridlines to drive attention to the bar heights.
- Dynamic sorting:** The X-axis automatically sorts occupations based on the total high-income proportion of the current selection, making top high earners identifiable.
- Strategic highlighting:** Viridis scale is perceptually uniform and provides better contrast for education levels.

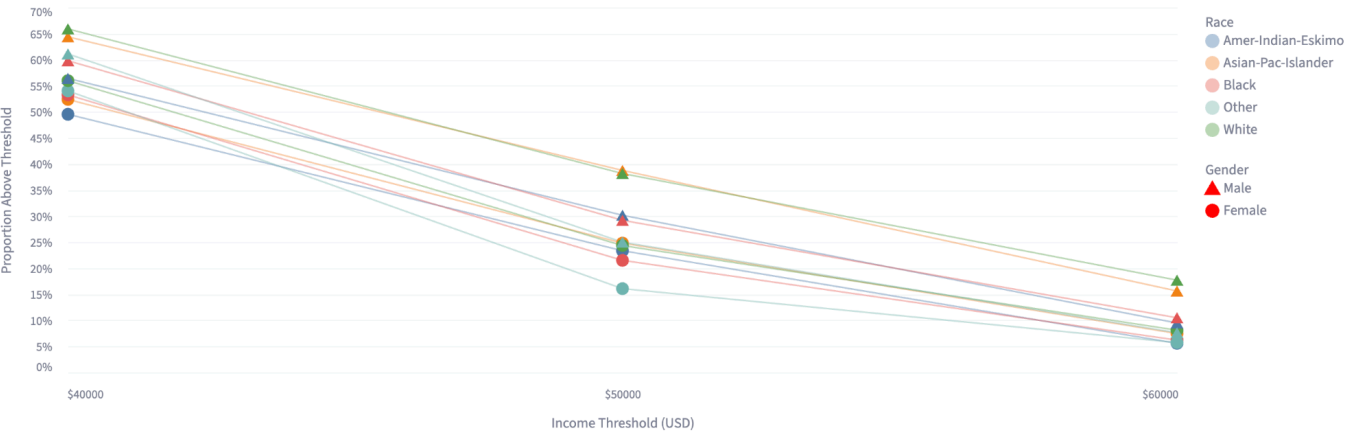
7. Multi-line chart

Demographics Exploration

How does changing the income threshold affect income disparity when comparing gender and race? 



Economic Access Disparity (40K - 60K)
Proportion of individuals exceeding income thresholds by race and gender



Maximizing Data-Ink Ratio: Removed chart borders and X-axis gridlines to focus attention entirely on the trend lines and data points.

Removing Clutter: Lower the Y-axis grid opacity and removed redundant axis domains for a cleaner view.

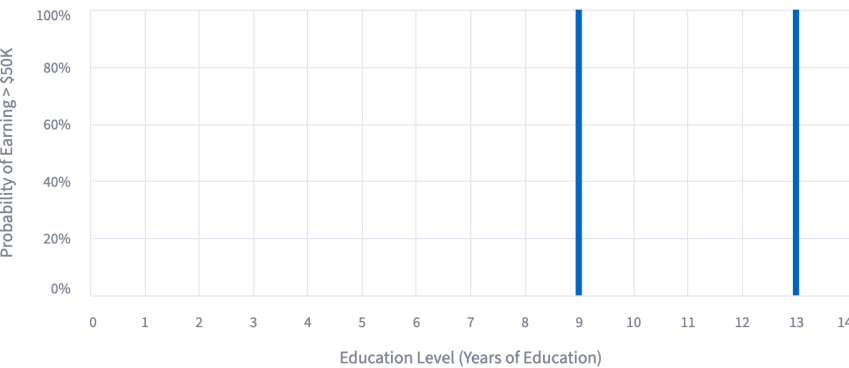
Strategic Highlighting: Tableau10 palette is better for color accessibility and increased point size to 160 for easier comparison of Gender (shape) and Race (color) .

LLM Visualizations

Prompt 1: Show the probability of earning more than \$50K by education level.

Your Visualization

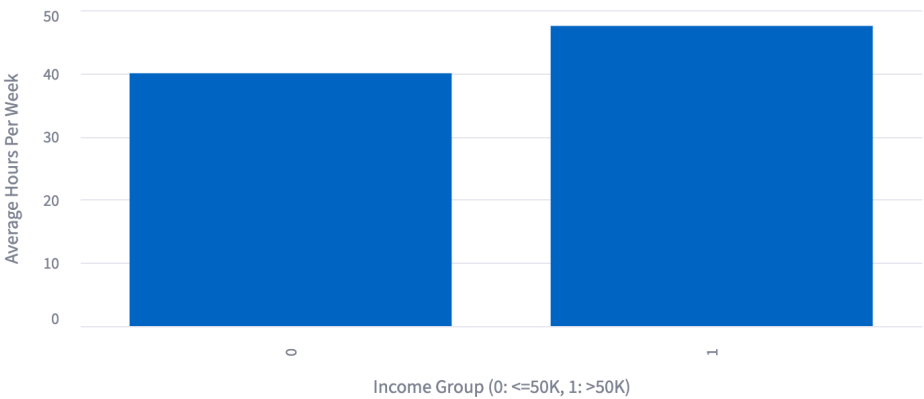
Probability of Earning More Than \$50K by Education Level



Prompt 2: Compare average hours worked per week between income groups.

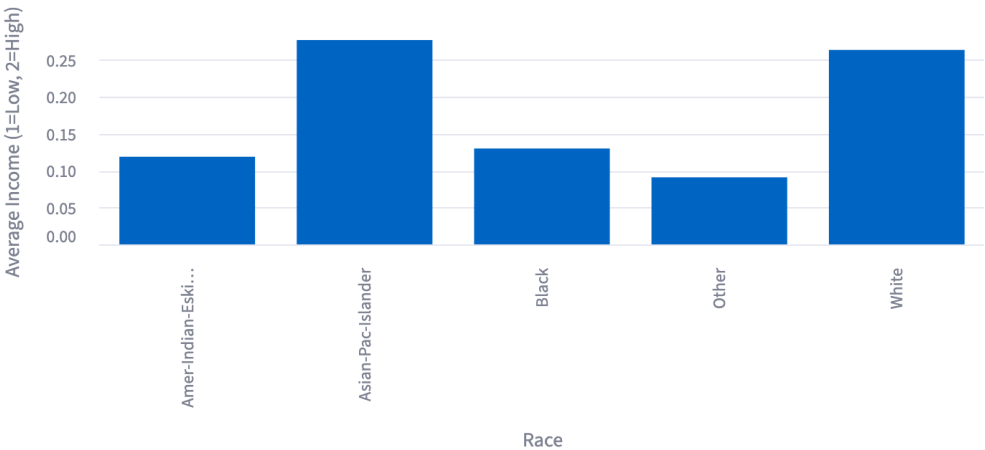
Your Visualization

Average Hours Worked Per Week by Income Group

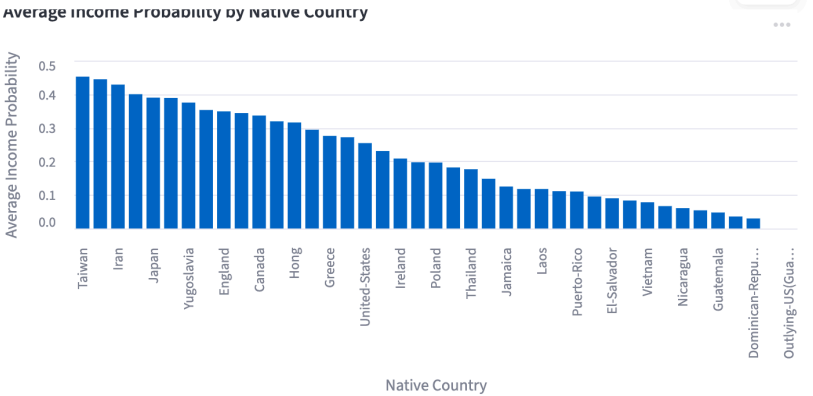


Prompt 3: Show how income probability differs across races.

Your Visualization



Prompt 4: Compare average income probability across native countries.



Cleaning and processing

Missing values

adult dataset: We decided only to use workclass, race, sex, native_country, income, education_num, and hours_per_week for this dataset. (**adult_new**):

- As noted in the metadata, the variables that have missing values are workclass and native_country.
- We drop the missing values of native_country and workclass since they only make up under 2% (583/32561) and just above 5% (1836/32561) of the dataset, dropping them would not compromise the statistical significance of the data set.

sup_countries dataset: We decided only to use population_density, literacy, agriculture, industry, and service for this dataset. (**countries_new**):

- As noted in the metadata, the variables that have missing values are agriculture, industry, and service.
- We replace all the nan (missing values) of these 3 variables with zeroes. We also assume that these missing values for each sector indicate that it has a very small to non-existent contribution to a country's GDP, or they weren't reported. While this approach might lower the global mean of these variables, we choose it to preserve the geographical diversity of our dataset.

Smoothing

adult_new: most of the columns we use for this dataset are categorical (race, workclass, native_country,...), so we cannot smooth them. Moreover, education_nums' values are discrete measurements, so smoothing them would turn them into decimals, this might add unnecessary values to the dataset.

countries_new: this data set is not a time series data, it records different countries at a single point in time (cross-sectional), therefore, smoothing is not needed for this dataset.

Aggregation

adult_new: We decided to aggregate the data set by the country because the adult data set contains individual data, but the country data set contains more general country data. We would not be able to meaningfully join the two tables without aggregating and take meaningful statistics to compare.

income_rate: Because of the conversion of income to a binary variable the mean represents the probability (or proportion) of individuals from that country earning more than \$50K.

education_num: Education is measured in years and we aggregated by the mean to show the average education level of groups individuals in the adult set that are from other countries.

hours_worked: The hours worked influences the income, and averaging the hours per country helps to understand whether certain immigrant groups work more hours or if income reflects the difficulty of jobs.

Post-join aggregation:

adult_new (region): We decided to aggregate the data set by the region because it gives a similar understanding of the data set as by country, but it does give a slightly different view allowing for deeper understanding of regional differences between immigrant groups.

Joins of input sources

We joined the adult data set with the country data set as well as the aggregated adult data set with the country data set. During the join, some of the countries in the adult data set do not exist in the countries data set, 7 of them to be specific. This is because they do not exist anymore or are considered parts of other countries. The unmerged entries will be excluded from any analysis involving the merged data set.